

How to Run Vanadis

1. Requirements

The version shown in the example run requires:

- GNU `make`,
- `gfortran` with OpenMP support,
- NVIDIA CUDA Toolkit (`nvcc`),
- `g++`,
- a compatible NVIDIA GPU and CUDA driver.

The current `Makefile` compiles the CUDA part for the following architectures:

```
sm_60
sm_61
sm_70
sm_75
```

The Fortran part is compiled with `-O3` optimization and OpenMP support. The CUDA and Fortran object files are then linked into the `vanadis` executable.

2. Compilation

From the directory containing the source code and `Makefile`, run:

```
make clean
make
```

`make clean` removes the previous build files:

```
cudatest.o
VANADIS.o
vanadis
```

A successful build creates:

```
vanadis
```

3. Setting the Number of CPU Threads

Vanadis uses OpenMP for the CPU part of the computation.

For example, to use four OpenMP threads under Linux:

```
export OMP_NUM_THREADS=4
```

The example run used four threads.

4. Running Vanadis

Start the model with:

```
./vanadis
```

At startup, Vanadis displays the current configuration parameters:

```
Q[kg/s]
K[m^2/s]
P[1/s]
V1[m/s]
V2[m/s]
V3[m/s]
ALPHA[m/s]
```

The example configuration was:

```
Q      = 1.0 kg/s
K      = 40.0 m^2/s
P      = 7.0E-4 1/s
V1     = 0.0 m/s
V2     = -5.0 m/s
V3     = 0.0 m/s
ALPHA  = 0.01 m/s
```

The program then asks:

```
IS IT CORRECT? [Y/N]
```

If the displayed parameters are correct, enter:

```
Y
```

Vanadis then starts the transient FEM calculation and writes the results to the `output` directory.

5. Calculation Output

For each time step Vanadis:

1. assembles the FEM matrix and vectors,
2. reports the matrix/vector assembly time,
3. solves the linear system,
4. reports the number of solver iterations and residual norm,
5. writes the solution to disk,
6. reports the total pollutant mass.

For example:

```
matrix/vector creation... nr of elements: 2166 nodes: 2800 time: 100 sec

solving a system of equations...

OK - number of iterations: 63
desired convergence tolerance: 0.1000E-07
residual norm: 0.8132E-08

czas=100
masa=96.403002800231789
```

6. 3-D Result Files

The main results are stored in files named:

```
T_ttttt.TXT
```

Each file contains:

```
X [m]
Y [m]
Z [m]
S(x,y,z) [mg/m^3]
```

that is, the complete three-dimensional concentration field for a given simulation time.

For the example calculation, output was produced for:

```
100 s
200 s
300 s
```

```
...  
2000 s
```

7. Automatic 2-D Cross-Sections

After the main calculation finishes, Vanadis automatically creates two-dimensional data files:

```
creating 2D data files...
```

The cross-sections are organized into three families:

```
X  
Y  
Z
```

corresponding to planes perpendicular to the three coordinate directions.

In the example run, the FEM calculation itself took approximately:

```
2.125 s
```

and generation of the 2-D files took approximately:

```
3.276 s
```

8. Creating a Ground-Level Bitmap with `VANADIS_BMP.exe`

After Vanadis has finished, the ground-level concentration field can be visualized with:

```
VANADIS_BMP.exe
```

The program is intended for Windows. Under Linux it can also be run using Wine.

For example, under Windows:

```
VANADIS_BMP.exe
```

or under Linux with Wine:

```
wine VANADIS_BMP.exe
```

The program creates a bitmap from:

```
X_02000_001.TXT
```

This file represents the **ground-surface concentration field at 2000 s**, corresponding to the approximately steady-state solution in the example simulation.

The resulting bitmap therefore provides a direct visualization of the near-ground pollutant concentration distribution.

The workflow is:

```
Vanadis
|
v
X_02000_001.TXT
|
v
VANADIS_BMP.exe
|
v
ground-level concentration bitmap
```

9. Optional Batch Generation of BMP Cross-Sections

A separate utility can also be used to generate bitmap images for the complete set of X/Y/Z cross-sections:

```
./bmp_xyz.exe output family
```

The argument:

```
output
```

specifies the directory containing the Vanadis results.

The argument:

```
family
```

selects **FAMILY** scaling, meaning that the same concentration scale is used for the same plane family and plane index across all simulation times.

In the example run, the utility found:

940

X/Y/Z files and successfully generated:

940/940

bitmap images.

Example filenames are:

```
X_00100_001.bmp  
X_00100_002.bmp  
...  
Z_02000_020.bmp
```

10. Complete Example Workflow

A typical Linux workflow is therefore:

```
make clean  
make  
  
export OMP_NUM_THREADS=4  
  
./vanadis
```

When Vanadis asks:

IS IT CORRECT? [Y/N]

enter:

Y

After the simulation has finished, a ground-level bitmap can be generated under Linux using Wine:

```
wine VANADIS_BMP.exe
```

or directly under Windows:

```
VANADIS_BMP.exe
```

The bitmap is generated from:

```
X_02000_001.TXT
```

which represents the ground surface at the approximately steady-state solution at $t = 2000 \text{ s}$.

Optionally, all available cross-sections can be converted to bitmap images with:

```
./bmp_xyz.exe output family
```

The complete workflow is therefore:

